

SensoPack 2026: Commercialization Roadmap



O2: 0.1%
TMA: <0.5ppm
STATUS: Nominal
SHELF-LIFE: +72hrs

Asset:
Smart Sensor-Embedded
Active Packaging

Target Market:
USD 16.2B
(Active & Intelligent)

Technology Readiness:
TRL 6 scaling to TRL 9

Classification:
Strictly Confidential

THE CORE ASYMMETRY OF FOOD WASTE

THE MACRO LOSS

**\$1
TRILLION**

1.3 billion tonnes of food lost annually across the global supply chain.

Perishable foods suffer the highest value loss, externalizing economic and environmental costs.

THE MICRO FAILURE

PET/PE Lidding Film

EVOH Barrier Layer

PP Tray Substrate

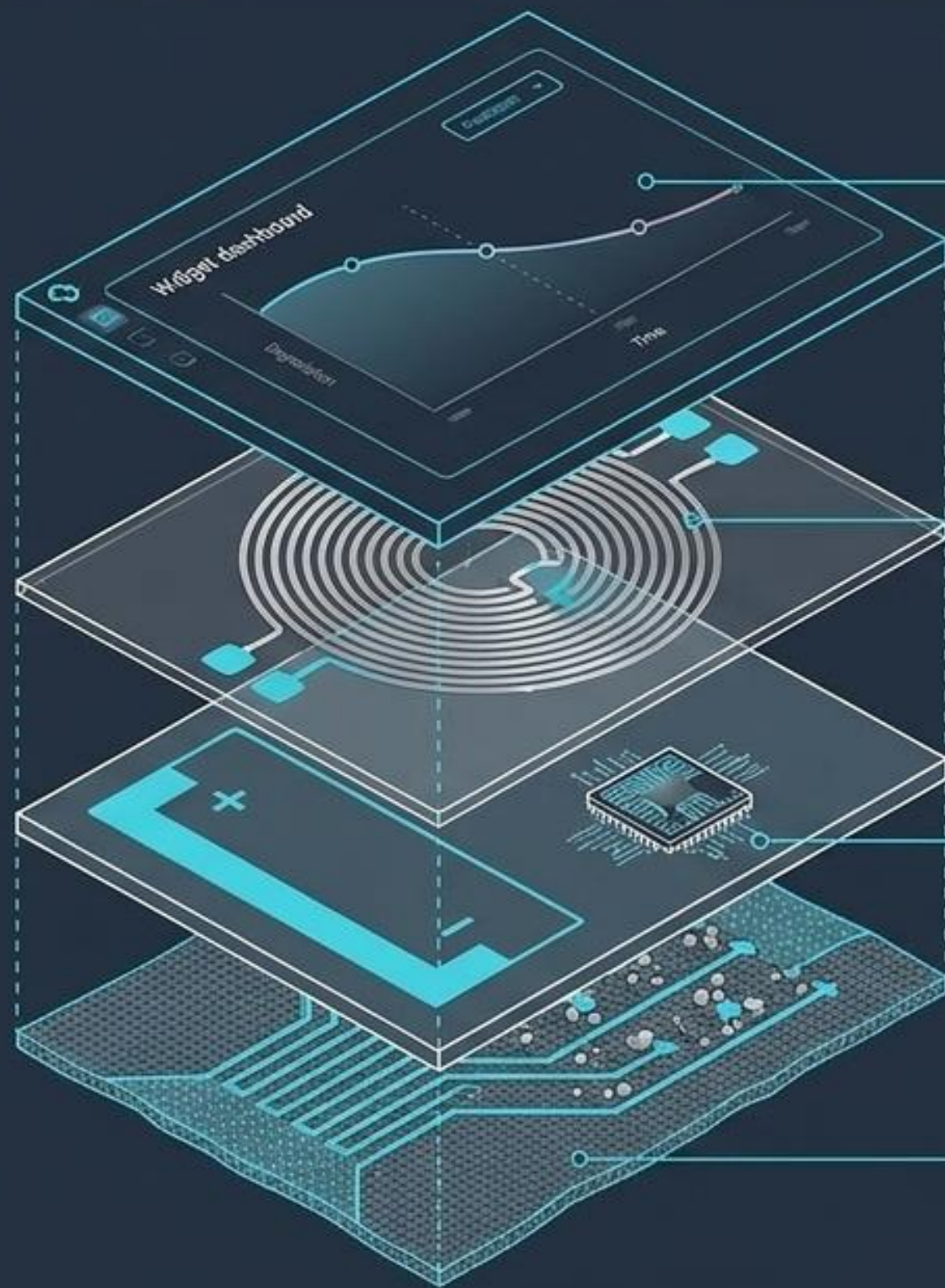
0.1–0.3% seal breach

Compromises the 20–80% CO₂ / 0–21% O₂ modified atmosphere, accelerating spoilage within 24–72 hours.

⚠ THE BLIND SPOT

Current visual inspections and random manual gas sampling **miss 70% of marginal failures** before they reach retail shelves. A fixed Best By label cannot account for this invisible physical reality.

The Functional Stack: Bridging Physics and Cloud AI



Layer 4: Cloud AI Inference

AWS/Azure Freshness Intelligence Platform applies Arrhenius-based degradation kinetics.

Layer 3: Communication

Silver nanowire NFC Type 2 antenna (13.56 MHz), oriented to bypass metalized barrier interference.

Layer 2: Power & Logic

Printed 1.2 mAh LiMnO₂ thin-film battery and 8x5mm 130nm CMOS ASIC.

Layer 1: Active Sensing

Silver nanoparticle and MnO₂ electrodes with Nafion membrane. MIP polymer detects TMA & biogenic amines.

Manufacturing Constraints Achieved

Total thickness added is <20 μm .

Operates seamlessly within standard hot-jaw sealing equipment.

The Analog VLSI Engine

Module 1: Electrochemical Front-End

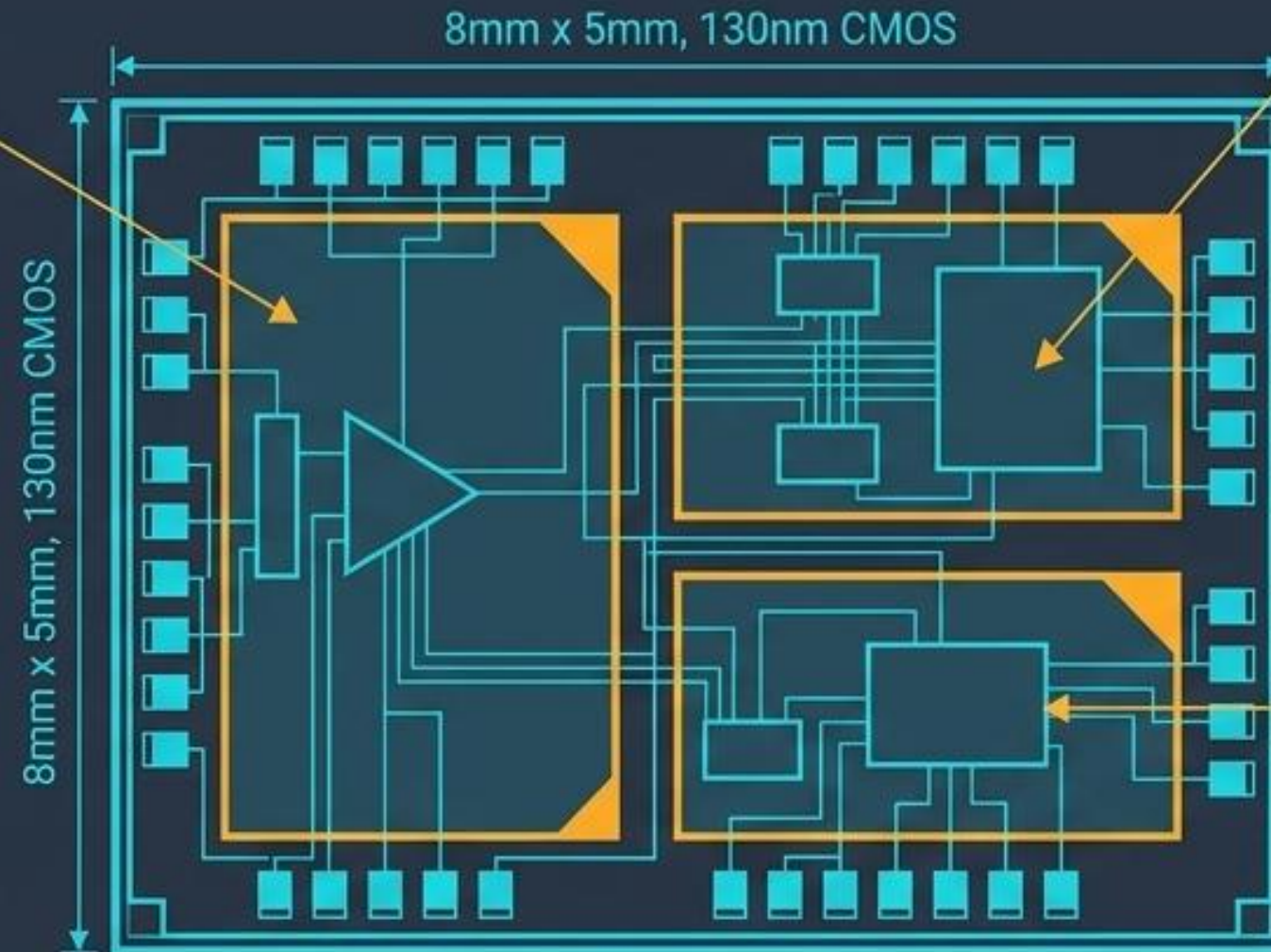
Potentiostat topology with on-chip transimpedance amplification. Handles 0.01%–21% O₂ detection with a T₉₀ <45 seconds.

Module 2: Impedance Spectroscopy

Switched-capacitor impedance bridge. Converts chemical binding events on the MIP layer into measurable impedance changes for ppm-level VOC detection.

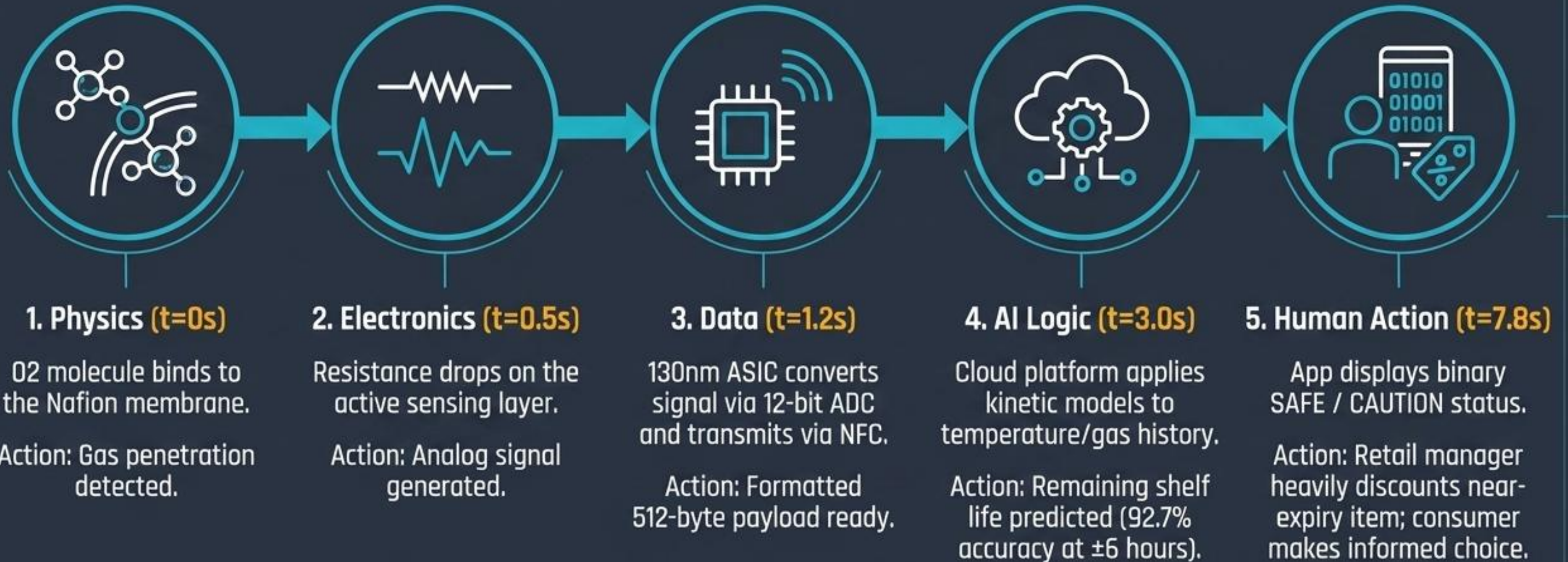
Module 3: Ultra-Low-Power Management

Sub-threshold CMOS bias circuits. Total current consumption <50 μ A. Ensures up to 730 days of operation at 20°C on a 1.2 mAh battery.



The Engineering Frontier: Integrating robust electrochemical sensing, 12-bit temperature-compensated ADC, and RF transceiver into a sub-200 μ m total package thickness.

The Real-Time Event Cascade



Commercial Reality: The Product Tier Strategy

Tier 1: Basic Traceability (Mass Market)

\$0.06 - \$0.24

Tech: Passive RFID
+ QR Code.
No battery.



Use Case: Dry goods, produce tracing,
basic inventory management.



Tier 2: Smart Package (Mid-Range)

\$0.36 - \$0.72

Tech: RFID + Temperature
sensor + Passive NFC.



Use Case: Retail dairy, mid-market
proteins, standard cold-chain.



Tier 3: Premium SensoPack (Industrial IoT)

\$1.80 - \$6.00

Tech: Multi-sensor (O₂,
VOC) + BLE/NFC +
Cloud AI Analytics.



Use Case: Export seafood, premium
meats, high-value pharmaceuticals.



This tiered architecture balances scalable unit costs with premium sensing capabilities, matching technology investment strictly to the value of the underlying perishables.

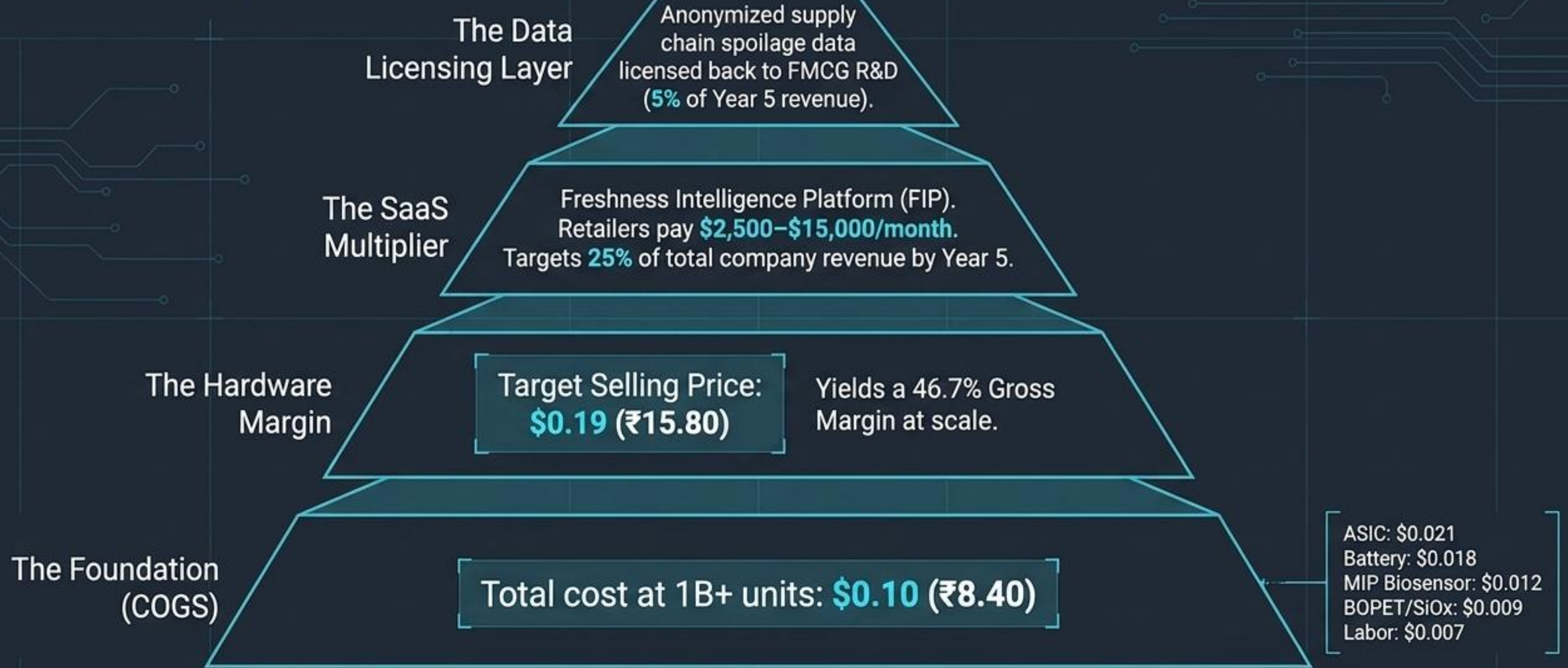
Competitive Diagnostic Matrix

	SensoPack	Senoptica	Blakbear	Standard RFID
O2 Sensing	● ● ● ● ●	● ● ● ●	● ●	○
VOC / Amine Sensing	● ● ● ● ●	○	● ● ●	○
NFC Consumer Readout	● ● ● ● ●	○	● ● ●	● ● ● ●
Integrated Lidding Film	● ● ● ● ●	● ● ● ●	○	○
AI Shelf-Life Prediction	● ● ● ● ●	● ●	● ● ●	○
Cold-Chain Traceability	● ● ● ● ●	● ●	● ● ●	● ● ● ● ●

Market Conclusion: SensoPack is the only TRL-6+ technology offering O2, VOC, and NFC in a single, inline-manufactured MAP laminate. Visual labels and mere data carriers fail the FSMA 204 mandate.

Capability	SensoPack™	Senoptica	Blakbear	TrackingPacking	Cellr	Hyperspectral Imaging
O ₂ sensing	●●●●●	●●●●○	●●○○○	○○○○○	○○○○○	●●●○○
VOC/biogenic amine sensing	●●●●○	○○○○○	●●●○○	○○○○○	○○○○○	●●●●○
NFC consumer readout	●●●●●	○○○○○	●●●○○	●●●●○	●●●●●	○○○○○
Integrated lidding film	●●●●●	●●●●○	●●○○○	●○○○○	○○○○○	N/A (external)
AI shelf-life prediction	●●●●●	●●○○○	●●●○○	●●○○○	○○○○○	●●●○○
Cost at scale (<USD 0.10/unit)	●●●●○	●●●●○	●●○○○	○○○○○	●●○○○	N/A
Regulatory approvals	●●○○○	●●●○○	●●○○○	●●●●○	●●○○○	●●●●●
Cold chain traceability	●●●●●	●●○○○	●●●○○	●●●●●	●○○○○	●●○○○

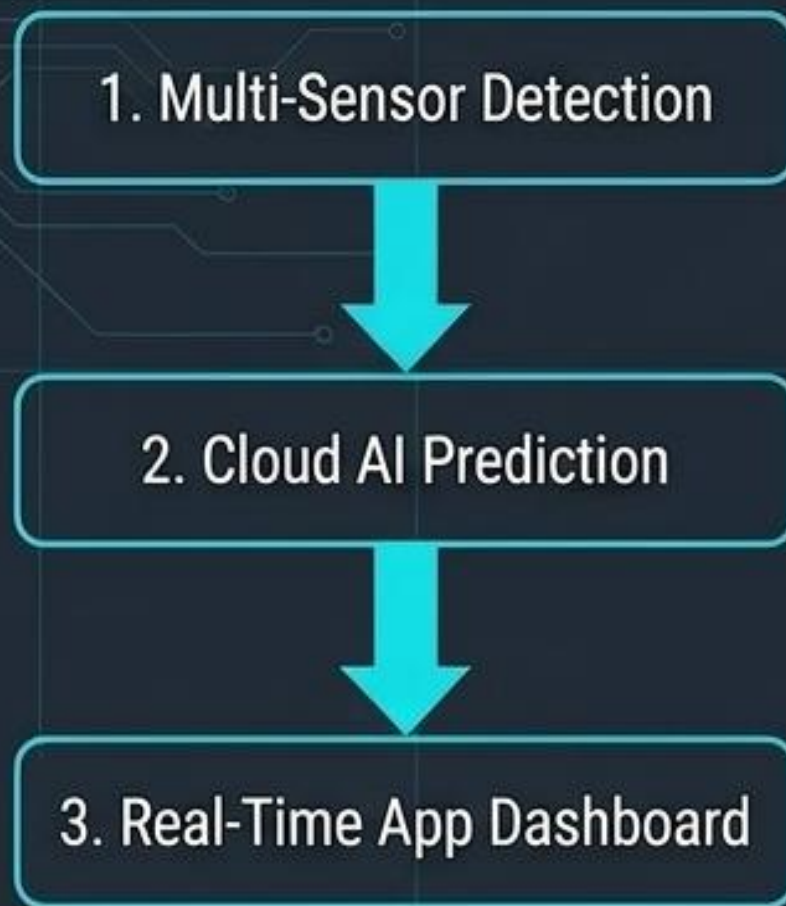
Unit Economics & The Profit Pyramid



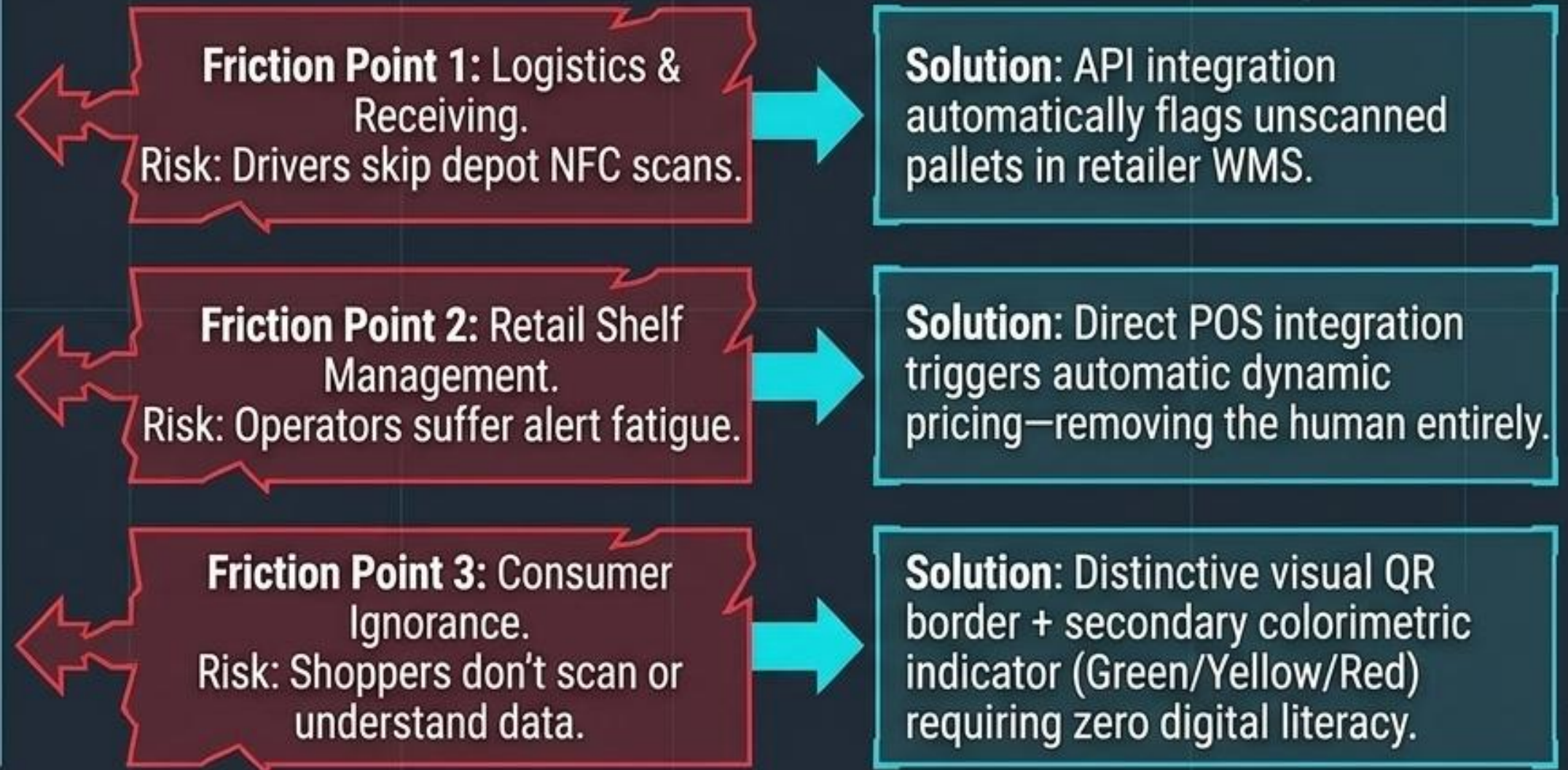
Financial Targets: Break-even at Month 34 post-launch. Year 5 Revenue target: **\$91.6M.**

The Behavioral Chasm: Solving for Human Friction

The Flawless Technical Flow

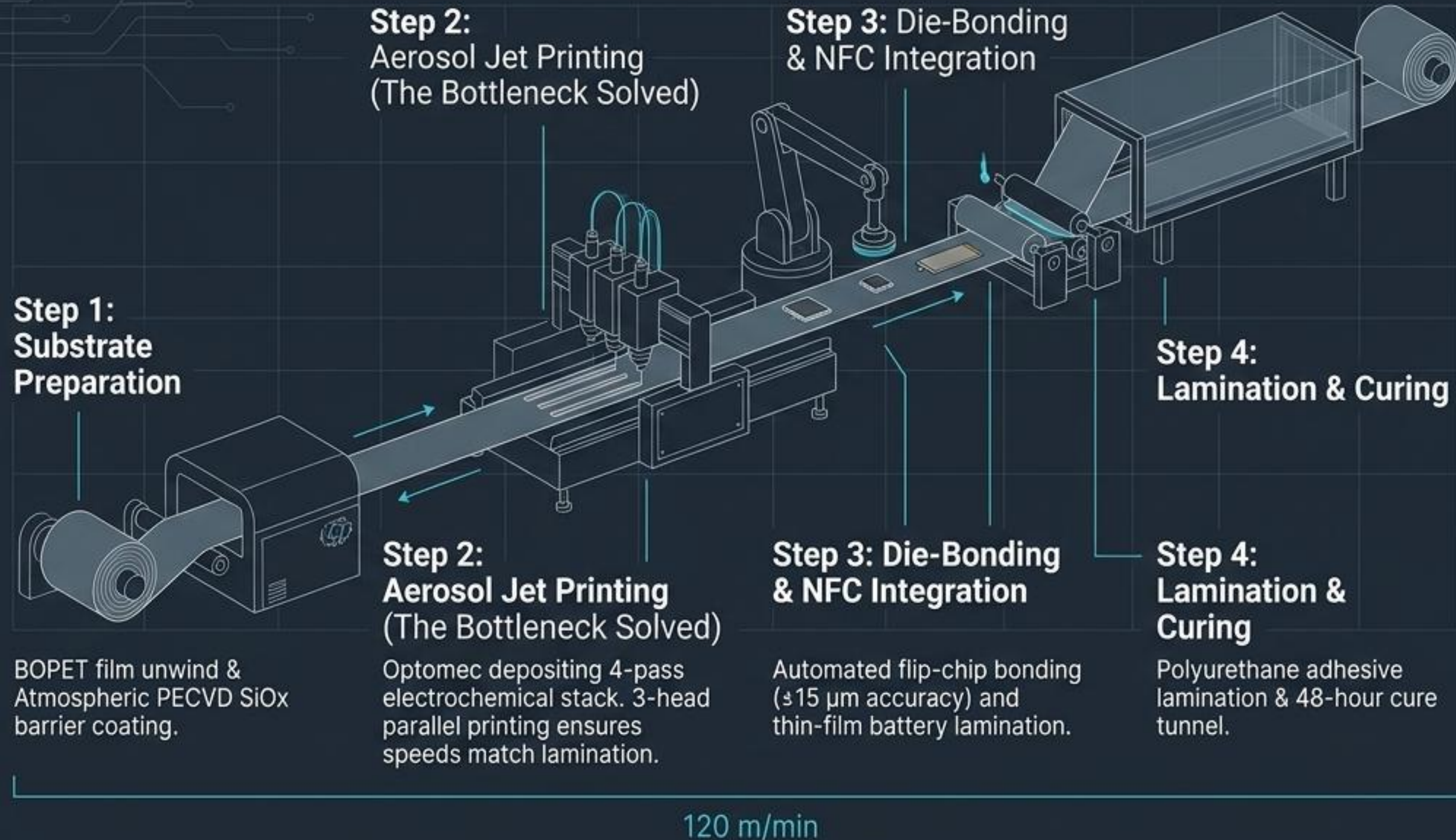


The Messy Human Flow



Core Go-To-Market Insight: We dedicate 30% of GTM budget to behavioral design, staff training, and system integration. Brilliant engineering has zero commercial value if the human ignores the data.

Manufacturing Integration & Scaling



Industry 4.0 Integration

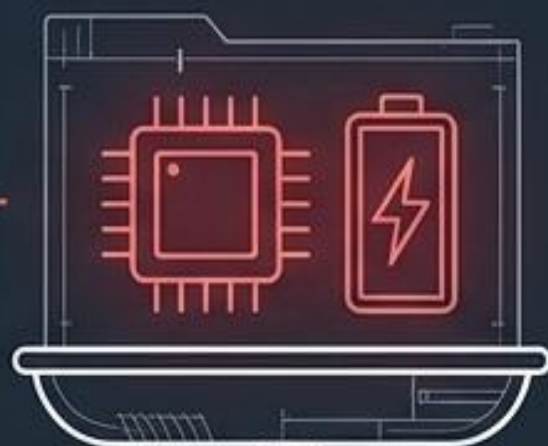
Real-time digital twin utilizes ML anomaly detection to identify process drift 2-4 hours before producing out-of-spec product. Targets <2.1% OEE loss.

The Net-Benefit Sustainability Equation

The Cost (Sensor LCA)

1,060g CO₂e footprint
per 1,000 units.

Driven primarily by **ASIC**
fabrication and **EoL**
battery disposal.



The Benefit (Avoided Food Waste)

4,200–8,600g CO₂e
saved per 1,000 units.

Prevents **3.5–7.2 kg** of
perishable food waste from
rotting in landfills.



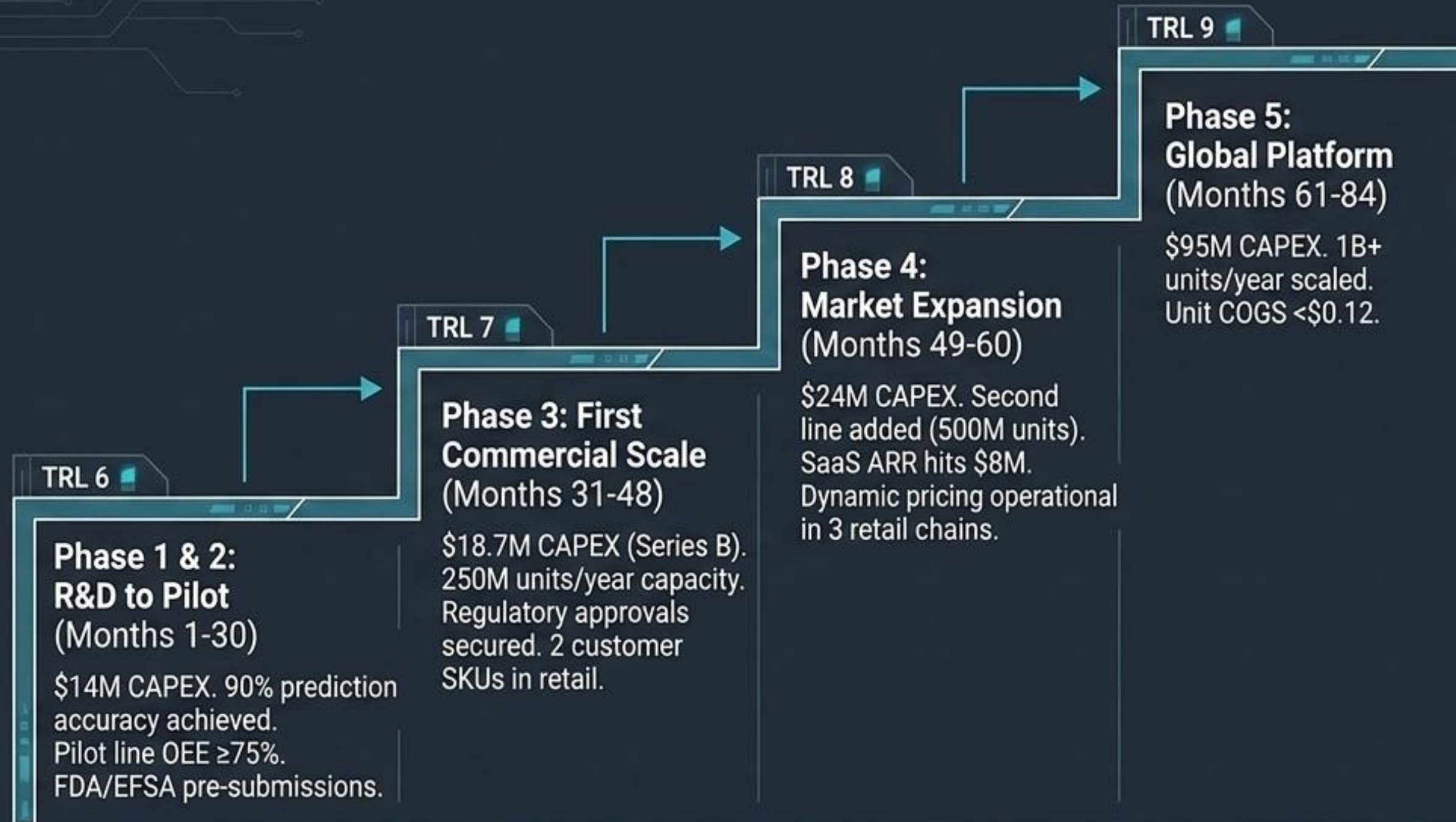
End-of-Life Loop: Electronics Take-Back Program

Consumers earn 'FreshPoints' for retail returns.

Target recovery rate: 35% by Year 3, 60% by Year 5.

**SensoPack delivers a net positive environmental benefit immediately,
validating at >11% food waste reduction effectiveness.**

7-Year Commercialization Roadmap & Milestones

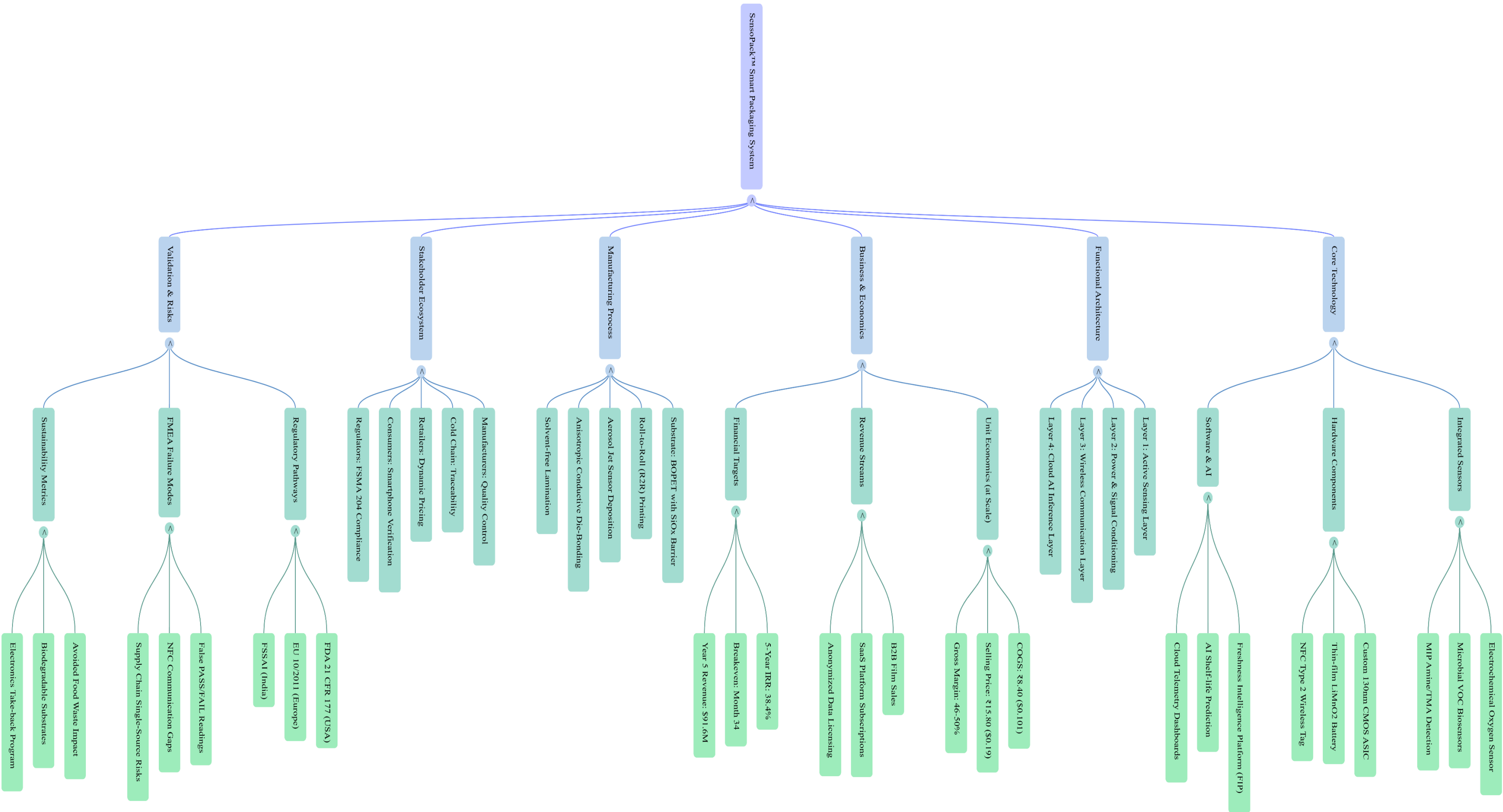


Financial Summary Box

Total Capital Requirement: \$167M over 84 months.

Base Case 5-Year IRR: 38.4%.

Exit Optionality: IPO (\$1.2B-\$2.2B valuation) or Strategic Acquirer (Amcor, Sealed Air) at 4-8x forward revenue.

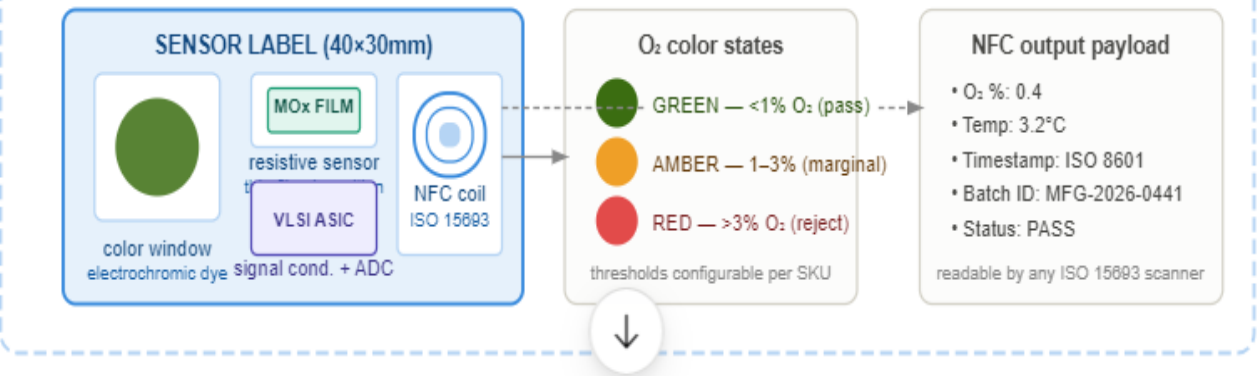


LAYER 1 — THE PHYSICAL SENSOR LABEL

What gets printed on the MAP lidding film

This is what a food manufacturer sees applied to their packaging line

MAP lidding film (packaging manufacturer applies label here)



LAYER 2 — THE PRODUCT SYSTEM

What the customer actually buys

Three components sold together as a qualified system

hardware

SMOSS Film Roll

Pre-printed sensor labels on compatible MAP lidding film. Drop-in replacement for existing film rolls. No line modification needed.

Format	Roll-to-roll, 330mm wide
Label pitch	Configurable per pack size
Shelf life (unopened)	18 months
Min order	50,000 units

software

SMOSS Cloud Platform

Web dashboard + REST API. Receives NFC scan events from line readers, tracks batch O₂ data, generates QC reports, and integrates with customer WMS/ERP.

Deployment	SaaS, EU data residency
API	REST + webhook
Compliance logs	ISO 22000, FSSC 22000
Pricing model	Flat subscription

hardware

Line Scanner Kit

ISO 15693 NFC reader, mounting bracket, and PoE module for inline installation. Reads every sealed pack at line speed and pushes to SMOSS cloud.

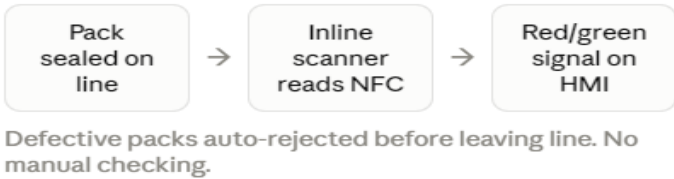
Read speed	>300 packs/min
Connectivity	Ethernet / Wi-Fi
Integration	MQTT / REST
Option	Handheld retail scanner

LAYER 3 — VALUE AT EACH POINT IN THE CHAIN

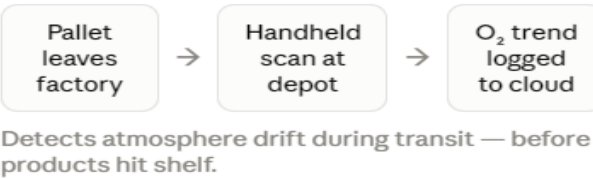
Who uses it and what they see

Each actor in the supply chain gets a different interface to the same underlying sensor

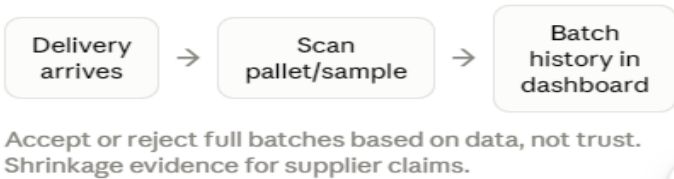
Factory QC operator



Logistics / cold chain



Retailer / buyer



Consumer (optional / phase 2)



LAYER 4 — COMMERCIAL PACKAGING (WHAT THE PROCUREMENT TEAM RECEIVES)

The commercial offer structure

Three tiers — pilot, standard, enterprise — each with different commitments and margins

PILOT

[Price TBD]

12-week proof of value

Volume	50K–200K units
Scanner kit	Loaned
Platform	Included
Support	Dedicated CSM

Customer pays cost-of-goods only. SMOSS absorbs margin to generate ROI data.

standard

[Price TBD]

Annual supply agreement

Volume	1M–10M units/yr
Scanner kit	Purchased
Platform	Per-seat SaaS
SLA	99.5% uptime

Primary commercial tier. Priced to deliver positive ROI at 0.4% shrinkage reduction.

ENTERPRISE

Custom

Multi-site / white label

Volume	>10M units/yr
Integration	Full ERP/WMS API
Data	On-premise option
Branding	White label available

Also covers licensing to packaging film OEMs — different P&L structure.

Component	Material	Technical Specification	Cost per Unit (USD)	Cost per Unit (INR) (Inferred)	Function	Risk Level
ASIC chip	Custom 130nm CMOS	8mm * 5mm, I ² C/SPI interface	0.0210	1.7430	Analog-to-digital conversion, temperature compensation, and data formatting	HIGH
Thin-film battery	LiMnO ₂ solid state	1.2 mAh; Operating range - 25 to +60°C	0.0180	1.4940	Powers the sensing circuit and NFC chip	Medium
MIP biosensor polymer	Custom MIP for TMA/biogenic amines	Selectivity coefficient >0.95	0.0120	0.9960	Detects spoilage biomarkers like trimethylamine and biogenic amines	HIGH
Substrate film	BOPET 12 μm	OTR <0.3 cm ³ /m ² /day	0.0095	0.7885	Flexible base layer for printed electronics	Low
Electrolyte membrane	Nafion® PFSA	1100 EW, 25 μm film	0.0085	0.7055	Proton-exchange polymer electrolyte for the electrochemical sensor	HIGH
Barrier coating	SiOx (silicon oxide CVD)	OTR contribution <0.5 cm ³ /m ² /day	0.0052	0.4316	Provides high oxygen barrier performance	Low
O ₂ working electrode	Silver nanoparticle ink	Particle size <50 nm, conductivity >5 * 10 ⁴ S/m	0.0042	0.3486	Electrochemical reduction of dissolved oxygen to generate current proportional to O ₂ pressure	Medium
NFC antenna ink	Silver nanowire conductive ink	Sheet resistance <5 Ω/sq	0.0038	0.3154	Forms the antenna for wireless data transmission (13.56 MHz)	Low
Reference electrode	MnO ₂ catalyst ink	Particle size <200 nm, purity >99.5%	0.0031	0.2573	Acts as reference electrode in the electrochemical sensor stack	Medium
Food-contact adhesive	Solvent-free polyurethane laminating adhesive	Compliant EU 10/2011, FDA 21 CFR	0.0028	0.2324	Bonds sensor-embedded film to structural lidding layers	Low

SensoPack™: The Future of Freshness Intelligence

The 4-Layer Intelligence Stack

Wireless NFC Communication

Smartphone-readable tags transmit timestamped freshness, temperature history, and 'Safe to Eat' status within 1.2 seconds.

ASIC Signal Processing

A custom 130nm CMOS chip converts chemical signals into digital freshness data at sub-50 μ A power.

Active Sensing & Power Layer

Embedded electrochemical O₂ sensors and thin-film batteries detect leaks and spoilage markers (TMA/Amines).



SensoPack's Digital Precision vs. Market Limitations

Traditional MAP

-  Expiry Tracking: **Static/Statistical**
-  Accuracy: **± 2-3 Days**
-  Consumer UX: **Visual Inspection**

SensoPack™ System

-  Expiry Tracking: **Dynamic/Real-Time**
-  Accuracy: **± 6-12 Hours**
-  Consumer UX: **Smartphone NFC Scan**

Impact & Economic Value



**35%-55%
Waste Reduction**

Significant reduction in retail food waste by replacing static 'Best By' dates with real-time accuracy.

±6-Hour Prediction Accuracy



Outperforms traditional printed labels (±3 days) using AI-calibrated degradation models across 47 food categories.



**₹8.40 (\$0.10)
Unit Cost**

Achieves mass-market scalability with a projected 36.4% IRR for high-volume commercialization.



Technology	Typical Unit Cost (High Volume)	Range	Battery	Reading Distance	Best Use
BLE (Bluetooth Low Energy) Beacon	₹100–400	Medium	Yes	10–100 m	Real-time monitoring

